

MOCK EXAMINATION — TRY BEFORE YOU BOOK

GCSE Biology

Mock Test & Mark Scheme

Time allowed	25 minutes
Total marks	41 marks
Number of questions	27
Topics covered	Cell Biology, Photosynthesis & Respiration, Genetics & Inheritance, Ecology

Student name: _____ **Date:** _____

Instructions

- Answer every question in the space provided on your own paper.
- Write your name and today's date at the top of your answer sheet.
- The marks for each question are shown in brackets, e.g. **(2 marks)**.
- Where a calculation is required, show your working — marks are available for method as well as the final answer.
- Once you finish, check your answers against the mark scheme at the back of this document.
- This mock test is provided free by BrainyEdHub to give you a feel for our teaching standard — book a free trial class to go through it with a tutor.

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Question Paper

Cell Biology

1. Name the organelle responsible for photosynthesis. [1 mark]
2. Name the organelle that controls a cell's activities. [1 mark]
3. State one function of the cell membrane. [1 mark]
4. Define 'diffusion'. [2 marks]
5. State two factors that affect the rate of diffusion. [2 marks]
6. Name the organelle where most aerobic respiration takes place. [1 mark]
7. State one difference between a plant cell and an animal cell. [2 marks]

Photosynthesis & Respiration

8. Write the word equation for photosynthesis. [2 marks]
9. Name the gas required for aerobic respiration. [1 mark]
10. Write the word equation for aerobic respiration. [2 marks]
11. State one difference between photosynthesis and respiration. [2 marks]
12. Name one factor that can limit the rate of photosynthesis. [1 mark]
13. State where in a plant cell photosynthesis takes place. [1 mark]

Genetics & Inheritance

14. Explain what is meant by a 'gene'. [2 marks]
15. A pea plant with genotype Tt is crossed with a plant with genotype tt. Using a genetic diagram, state the expected ratio of Tt : tt offspring. [3 marks]
16. Explain what is meant by a 'dominant' allele. [2 marks]
17. Explain what is meant by a 'recessive' allele. [2 marks]
18. Define 'genotype'. [1 mark]
19. Define 'phenotype'. [1 mark]
20. State whether human body cells are haploid or diploid. [1 mark]

Ecology

21. Define 'ecosystem'. **[1 mark]**
22. Give an example food chain with three organisms, showing the direction of energy flow. **[2 marks]**
23. State one human activity that can reduce biodiversity, and explain its effect. **[2 marks]**
24. Define 'habitat'. **[1 mark]**
25. Name the process by which energy is lost between trophic levels in a food chain. **[1 mark]**
26. State one adaptation that helps a polar animal survive in cold conditions. **[1 mark]**
27. Explain what is meant by 'competition' between organisms. **[2 marks]**

Mark Scheme

Use this guide to check your answers. Award full marks for correct final answers; where working is shown, award partial credit as indicated.

Q	Model answer / mark scheme	Marks
1	Chloroplast	1
2	Nucleus	1
3	It controls which substances enter and leave the cell.	1
4	The net movement of particles from an area of higher concentration to an area of lower concentration, down a concentration gradient.	2
5	Any two of: concentration gradient, temperature, surface area.	2
6	Mitochondria	1
7	E.g. plant cells have a cell wall, chloroplasts and a large permanent vacuole; animal cells do not.	2
8	Carbon dioxide + water → glucose + oxygen (using light energy).	2
9	Oxygen	1
10	Glucose + oxygen → carbon dioxide + water (+ energy).	2
11	Photosynthesis stores energy and only happens in green plants/algae in light; respiration releases energy and happens in all living cells, all the time.	2
12	E.g. light intensity, carbon dioxide concentration, or temperature.	1
13	In the chloroplasts.	1
14	A short section of DNA that codes for a particular protein or characteristic.	2
15	1 : 1 Tt × tt gives offspring Tt, Tt, tt, tt.	3
16	An allele whose characteristic appears in the offspring even if only one copy is inherited.	2
17	An allele whose characteristic only appears in the offspring if two copies are inherited.	2
18	The genetic makeup of an organism for a particular characteristic.	1
19	The observable characteristic that results from an organism's genotype.	1
20	Diploid	1
21	All the organisms living in a particular area, together with their non-living environment.	1

Q	Model answer / mark scheme	Marks
22	E.g. Grass → Rabbit → Fox (arrows point from the organism being eaten to the one eating it).	2
23	E.g. Deforestation destroys habitats, reducing the number of species an area can support.	2
24	The natural environment in which an organism lives.	1
25	Respiration (energy is lost as heat).	1
26	E.g. a thick fur coat or a layer of blubber for insulation.	1
27	When two or more organisms compete for the same limited resource, such as food, space, water or light.	2
Total for Biology		41 marks

How did you do? A BrainyEdHub tutor can go through any question you got stuck on, and build a personalised revision plan around your results. Book a free trial class at brainyedhub.com/contact.html or message us on WhatsApp.